IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

SpaceX focuses on reducing the cost of space missions through reusable rocket technology, particularly the Falcon 9 first-stage boosters. The success of landing these boosters is crucial for improving cost efficiency and operational sustainability, but outcomes vary due to factors such as payload mass, orbit type, launch site, and booster configuration. Leveraging historical launch data, this project applies data science techniques—including data collection, preprocessing, exploratory analysis, and machine learning—to identify patterns and predict landing success. The goal is to generate data-driven insights and build predictive models that can support more informed decision-making in space launch operations.

Introduction

Project Background:

•SpaceX focuses on reducing space mission costs through **reusable rocket technology**, especially Falcon 9 boosters.
•Successful landing of the first-stage booster plays a critical role in **cost efficiency and mission sustainability**.
•However, landing outcomes vary due to factors such as **payload mass, orbit type, launch site, and booster version**.
•Using historical launch data, this project applies **data science techniques and machine learning models** to analyze patterns and predict landing success.
•The objective is to generate **data-driven insights and predictive capabilities** to support better decision-making in launch operations

Introduction

Problem Statements :

1. Can we accurately predict the success of SpaceX Falcon 9 first-stage landings using historical launch data?
2. Which key factors (payload mass, orbit type, launch site, booster version) most significantly influence launch success?
3. What patterns and trends can be identified in SpaceX launch data through exploratory data analysis and SQL?
4. How can interactive visualizations (dashboards and geospatial maps) to enhance understanding of launch performance and outcomes?
5. Which classification model performs best in predicting launch success, and how can hyperparameter tuning improve its accuracy?

Section 1

Methodology

Methodology

Executive Summary

- Data collection methodology:
  - The data was collected using the SpaceX Rest API and web scraping, where JSON data was extracted, converted to tabular format and integrated into a unified dataset.
- Perform data wrangling
  - The data was processed, including handling missing values, cleaning and transforming features to prepare a structured dataset for further analysis.
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
  - Classification models were built by training on split data, tuned using GridSearchCV, and evaluated using accuracy and confusion matrix to select best model.

Data Collection

Key Phrases :

- SpaceX Rest API
- HTTP Requests
- JSON Data Extraction
- Data Parsing and Normalization
- Web Scraping (HTML Parsing)
- Data Integration

Data Collection – SpaceX API

Adding the GitHub URL of the completed SpaceX API calls notebook [SpaceX Data Collection Notebook] as an external reference and peer-review purpose

SpaceX API
↓
Request
↓
JSON
↓
Parse Structure
↓
Feature Engineering
↓
Dataset
↓
Analysis

Data Collection - Scraping

Adding the GitHub URL of the completed web scraping notebook [SpaceX Web Scraping Notebook] as an external reference and peer-review purpose

Web Page (URL)
↓
HTTP Request (requests)
↓
HTML Content
↓
Parsing (BeautifulSoup)
↓
Data Extraction (tags, tables)
↓
Data Cleaning (re, unicodedata)
↓
Data Structuring (pandas)
↓
Final Dataset
↓
EDA / Visualization / ML

Data Wrangling

Key Phrases :

- Raw Data Ingestion
- HTML Parsing (BeautifulSoup)
- Data Extraction (mission, rocket, payload, outcome)
- Missing Value Handling
- Text Cleaning (re)
- Unicode Normalization
- Data Transformation
- Data Type Conversion
- Feature Engineering
- Data Structuring (pandas DataFrame)
- Data Validation

EDA with Data Visualization

Charts Used and Why:

- **Boxplot** – Used to visualize the distribution of numerical variables and detect outliers.
- **Regplot (Regression Plot)** – Used to examine relationships and trends between two numerical variables, showing correlation.
- **Heatmap** – Used to display correlations between multiple variables, helping identify strong or weak relationships at a glance.

Github Link - EDA with Data Visualisation

EDA with SQL

Features Bullet Points:

- **SELECT:** Chose relevant columns from datasets.
- **WHERE:** Filtered rows by neighborhood, crime type, or school.
- **GROUP BY:** Aggregated counts, averages, sums.
- **ORDER BY:** Sorted for top/bottom values.
- **JOIN:** Combined datasets for cross-analysis.
- **HAVING:** Filtered aggregated results.
- **DISTINCT:** Retrieved unique values.
- **LIMIT:** Restricted output to top N records.

Github Link - [EDA-with-SQL-Notebook]

Build an Interactive Map with Folium

Map Objects Created and Added

- **Markers & MarkerClusters:** Placed multiple points on the map; clusters reduce clutter when many markers are close together.
- **CircleMarkers:** Represented magnitude or intensity of data at specific locations.
- **FeatureGroups:** Grouped related markers or layers for better organization and layer control.
- **Lines / PolyLines:** Showed connections or routes between points (if applicable).
- **Popups & Tooltips:** Displayed interactive information on click or hover for user engagement.

Purpose

- These objects enhance **interactivity and visual clarity**, allowing users to **explore patterns, trends, and key locations** on the map efficiently, while **organizing layers and reducing visual clutter**.

GitHub URL - Interactive Map using Folium.

Build a Dashboard with Plotly Dash

Summarize what plots/graphs and interactions have added to a dashboard:

1. Recession Period Statistics

- **Line Chart**: Avg automobile sales over years (during recession)
- **Bar Chart**: Avg sales by vehicle type
- **Pie Chart**: Advertising expenditure share by vehicle type
- **Bar Chart**: Impact of unemployment rate on sales by vehicle type

2. Yearly Statistics

- **Line Chart**: Overall yearly sales trend
- **Line Chart**: Monthly sales trend
- **Bar Chart**: Avg sales for selected year
- **Pie Chart**: Ad expenditure by vehicle type for selected year

Explain why you added those plots and interactions

Interactions Added

- **Dropdown 1**: Select report type (Yearly vs Recession)
- **Dropdown 2**: Select year (enabled only for Yearly Statistics)
- **Dynamic display**: Charts update based on selections

Why These Were Added

- **Trend analysis** (line charts): To observe sales patterns over time
- **Comparison** (bar charts): To compare performance across vehicle types
- **Proportion insight** (pie charts): To show share of ad spending
- **Economic impact**: To understand how recession & unemployment affect sales
- **User control** (dropdowns): To make the dashboard interactive and focused

GitHub URL - [Creating-Dasboard-using-Plotly-Dash]

Predictive Analysis (Classification)

- Built classification models using Logistic Regression and Random Forest on preprocessed data with Gaussian noise to simulate real-world errors, evaluated performance using accuracy, precision, recall, F1-score, and confusion matrix with focus on minimizing false negatives, improved models through hyperparameter tuning using GridSearchCV and cross-validation, and selected Random Forest as the best-performing model due to its higher recall and F1-score and better robustness to noise.

- Performed feature scaling and proper train-test split to prevent data leakage and ensure reliable model evaluation.

- Conducted comparative analysis of model performance and validated stability across folds to ensure strong generalization

Key Phrases and Flowchart

- Cross Validation
- Model Optimization
- Bias-Variance Tradeoff
- Performance Benchmarking

Github URL -[Machine-Learning-Predictive-Analysis-Classification]

Start
↓
Load Dataset
↓
Add Gaussian Noise
↓
Train-Test Split
↓
Feature Scaling
↓
Model Training
↓
Hyperparameter Tuning
↓
Cross Validation
↓
Model Evaluation
↓
Confusion Matrix Analysis
↓
Model Comparison
↓
Best Model Selected
↓
End

Results

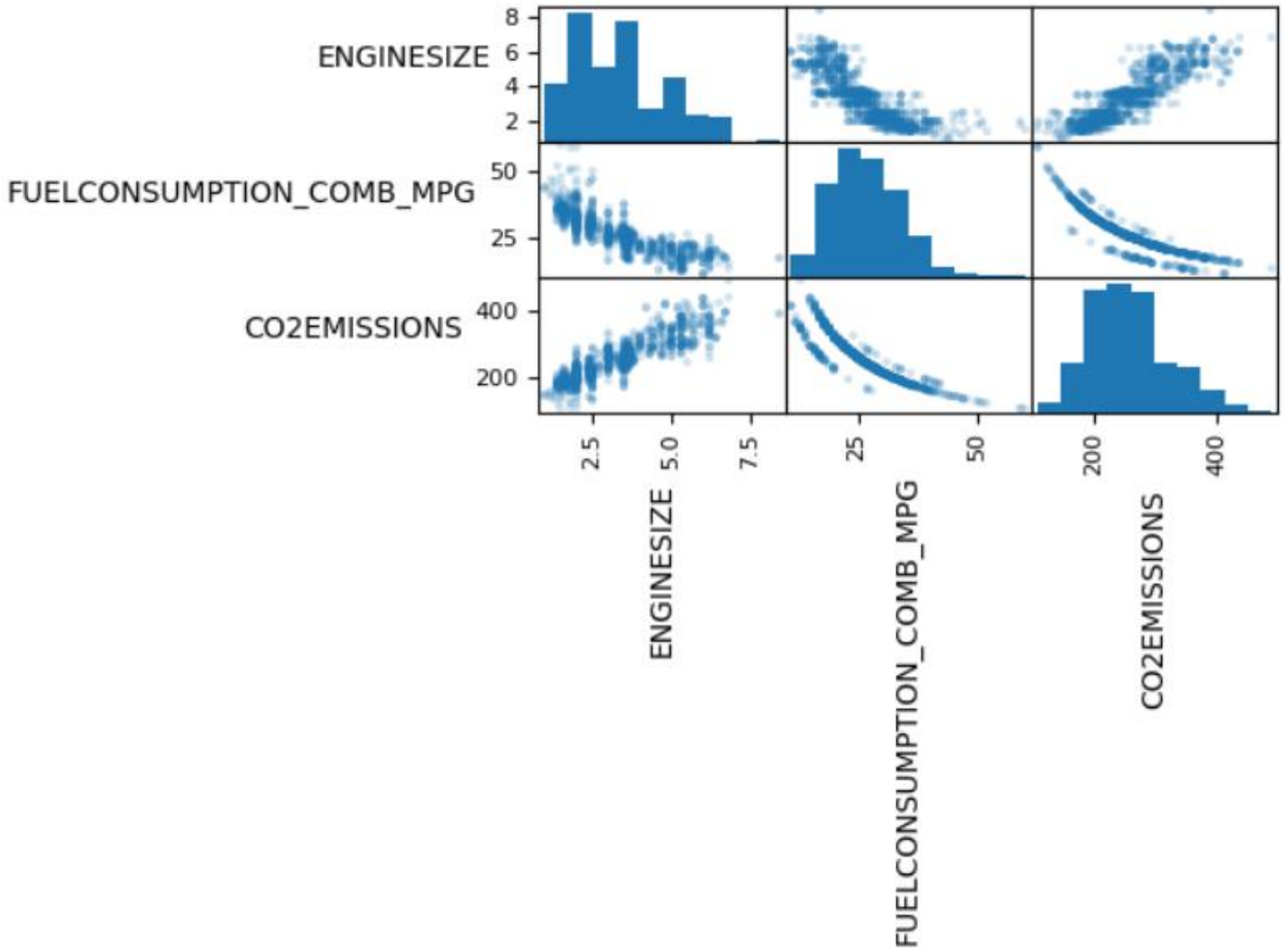
Exploratory data analysis results

- Exploratory Data Analysis of laptop pricing revealed that features like **brand, RAM, processor type, and storage** have a strong influence on price variations, with premium brands and higher specifications driving higher costs.
- Correlation and distribution analysis helped identify key price drivers and patterns, enabling better feature selection and more accurate predictive modeling.

Predictive analysis results

- The model accurately predicts whether a tumor is benign or malignant, with **Random Forest** achieving higher recall and minimizing false negatives, which is critical for early breast cancer detection.
- Even with noisy data, the optimized model remained robust, ensuring **reliable predictions** in real-world medical scenarios.

Interactive analytics demo in screenshots



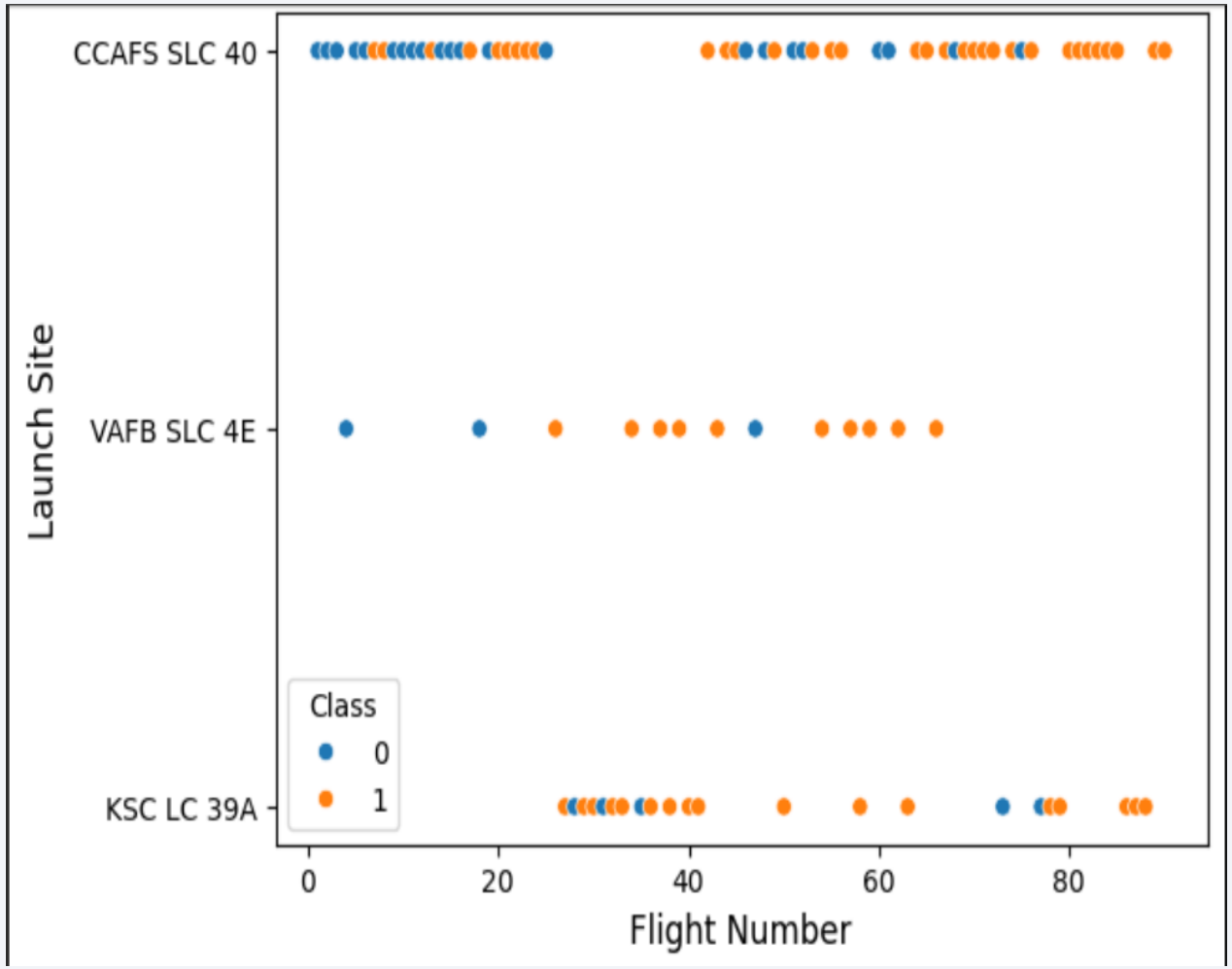
Section 2

Insights drawn from EDA

Flight Number vs. Launch Site

Showing the screenshot of the scatter plot with explanations:

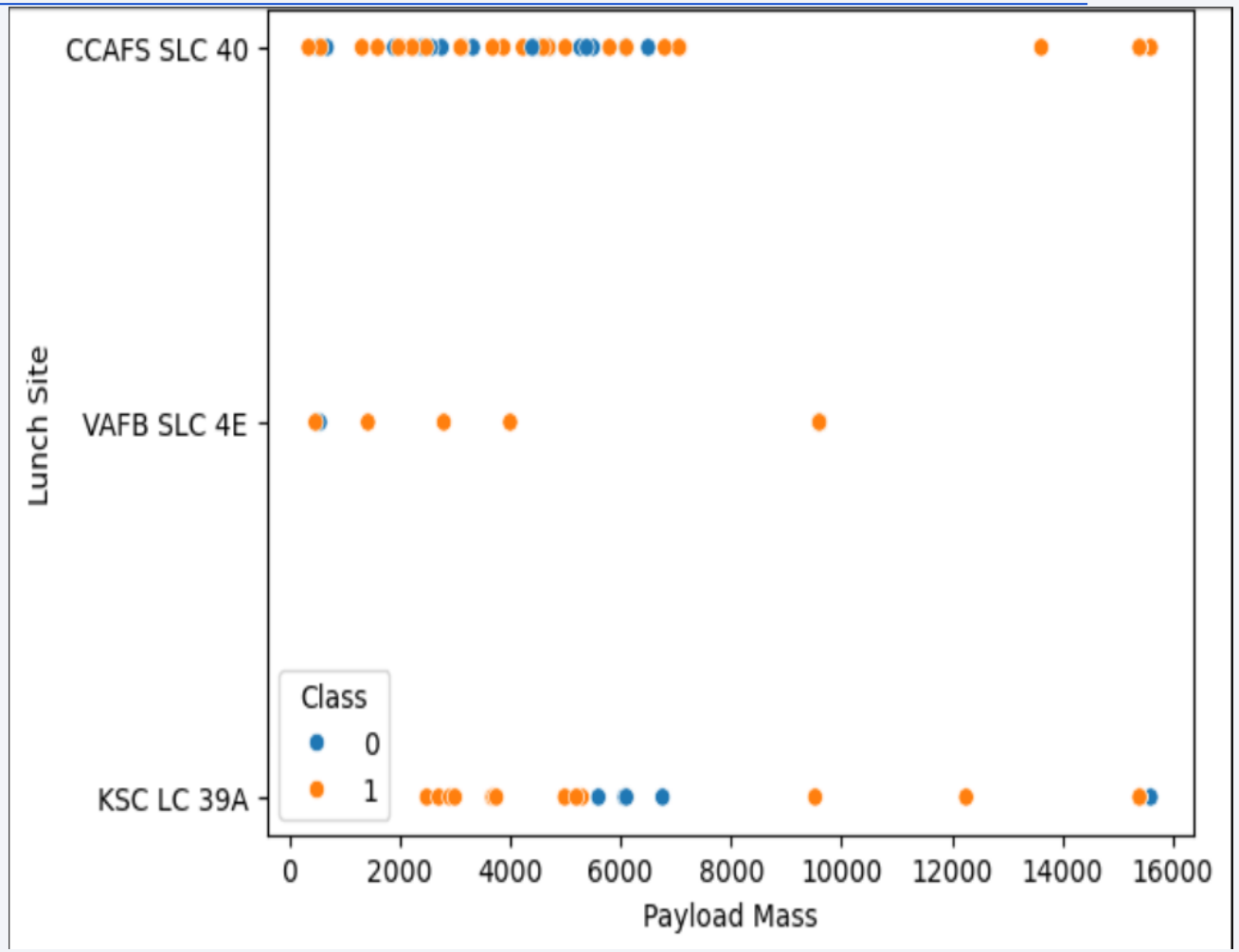
- Some sites (like **CCAFS SLC 40**) have **more launches overall.**
- Higher concentration of orange dots means **higher success rate.**
- Scattered blue dots shows **failures at different points.**
- Launch success seems to **increase with experience (higher flight numbers)**



Payload vs. Launch Site

Showing the screenshot of the scatter plot with explanations:

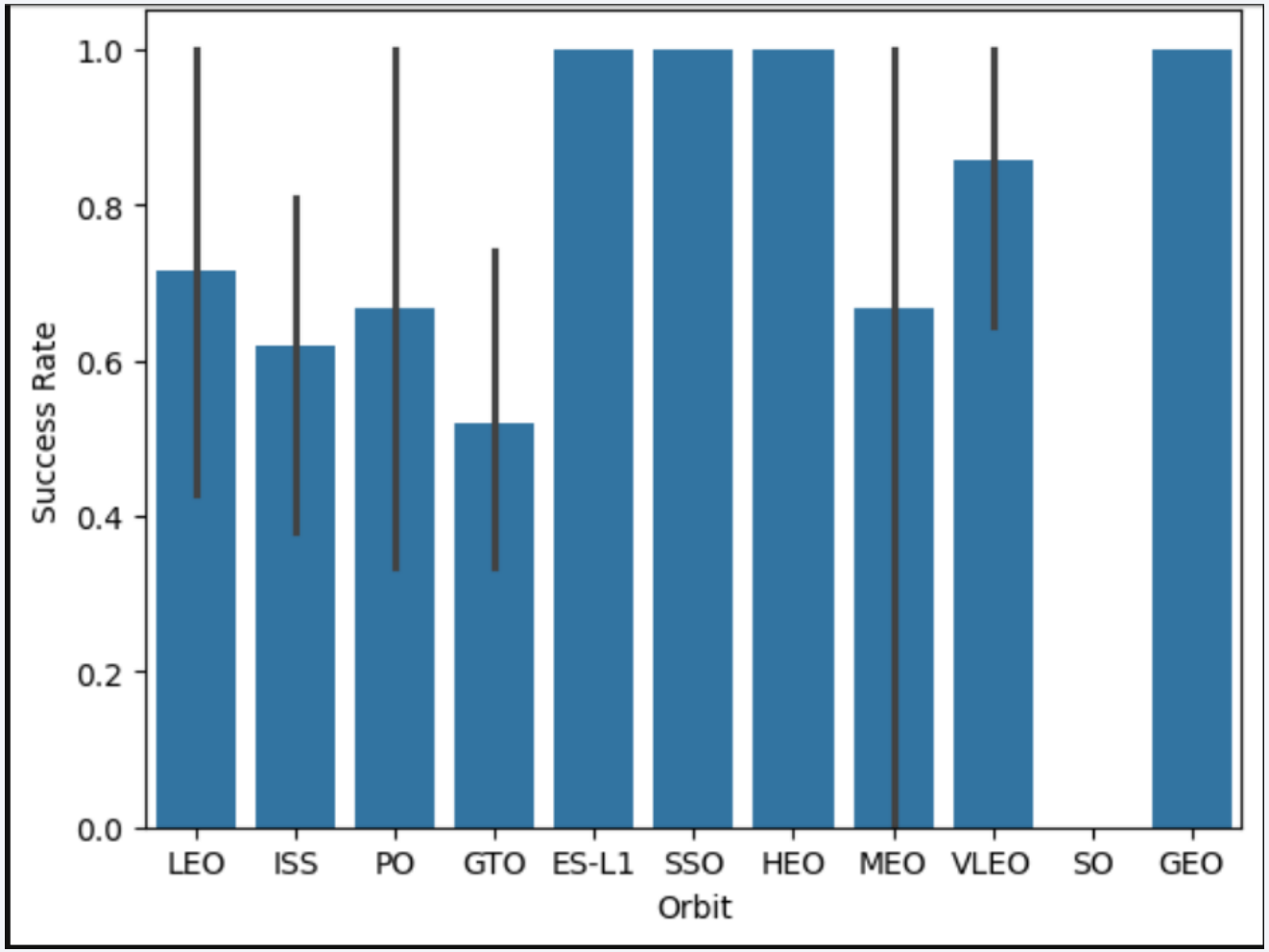
- Most launches across sites are **successful (more orange dots).**
- Payloads vary widely, especially at **CCAFS SLC 40.**
- Higher payloads show **good success rates.**
- Launch success depends more on **other factors than just payload weight.**



Success Rate vs. Orbit Type

Showing the screenshot of the scatter plot with explanations

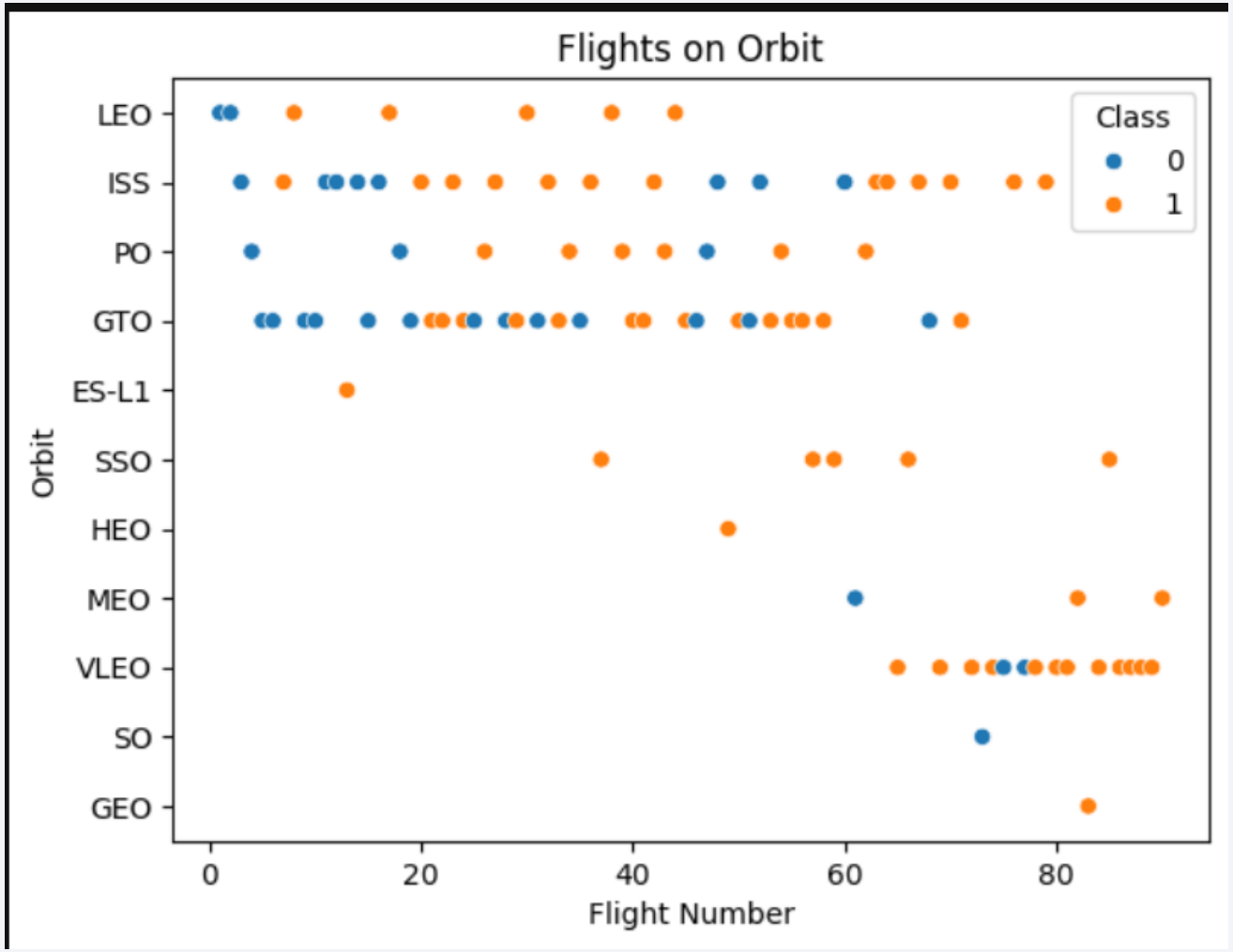
- Some orbits (like ES-L1, SSO, HEO, GEO) have **very high success rates (close to 1).**
- Variability differs across orbits, indicating **inconsistent performance in some cases.**
- Certain orbits are more reliable, while others are more challenging and show **lower or variable success rates**



Flight Number vs. Orbit Type

Showing the screenshot of the scatter plot with explanations.

- Most launches across orbits are **successful (more orange points).**
- Orbits like **SSO, HEO, GEO** show mostly successful launches.
- Later flight numbers tend to have **higher success concentration.**
- Performance improves over time, with more successful missions in later flights



Payload vs. Orbit Type

Showing the screenshot of the scatter plot with explanations:

•**GTO**: strongly dominated by orange (Class 1), masses mostly 3,000–6,500 kg.

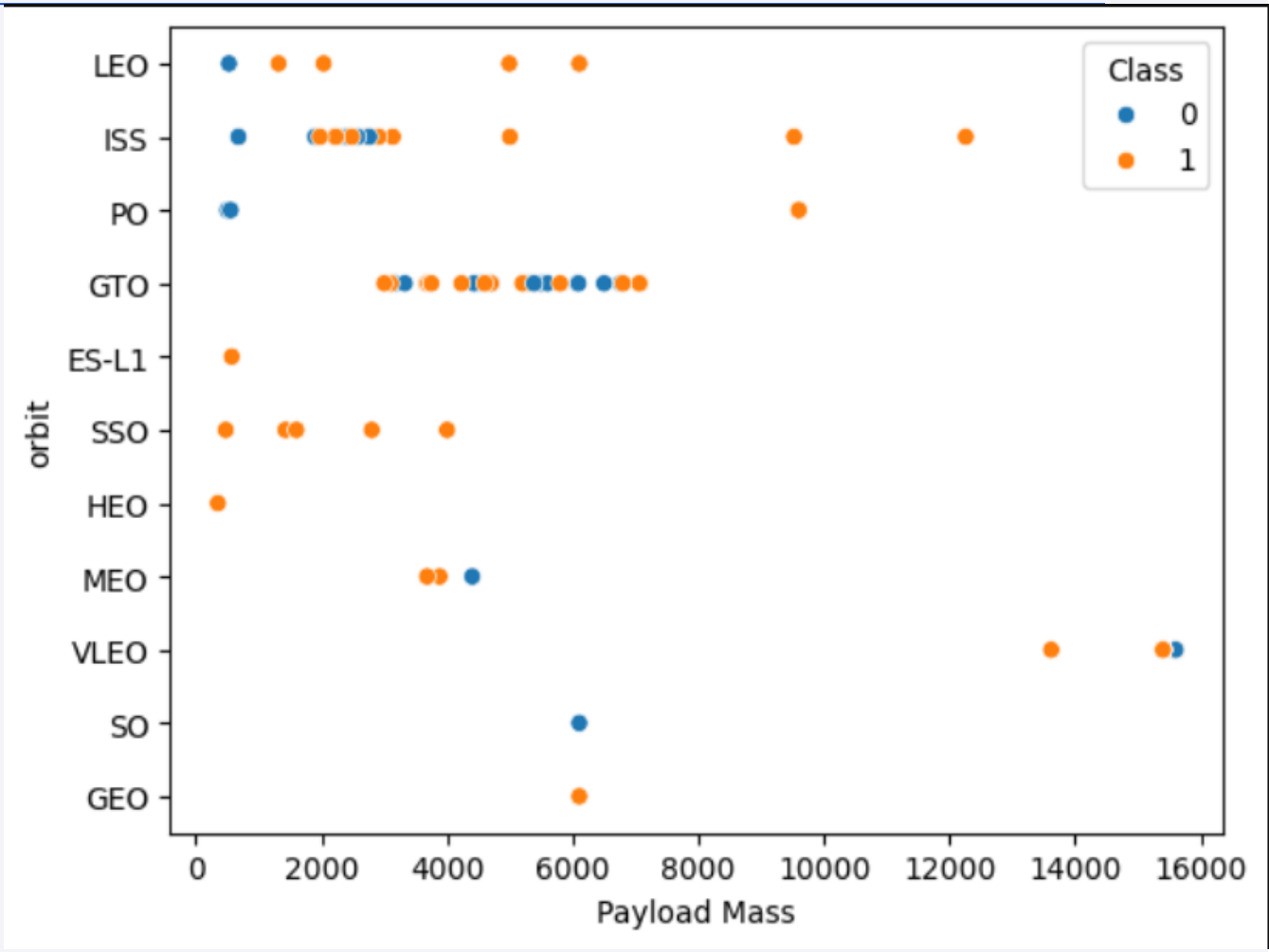
•**GEO**: almost only orange, very heavy payloads (~6,000+ kg) .

•**SSO**: mostly orange, typically lighter payloads (< ~4,000 kg) .

•**LEO & ISS**: very mixed (blue + orange), extremely wide mass range (100–16,000+ kg)

•**MEO, HEO, ES-L1**: sparse, mostly medium masses, mixed or slightly more blue.

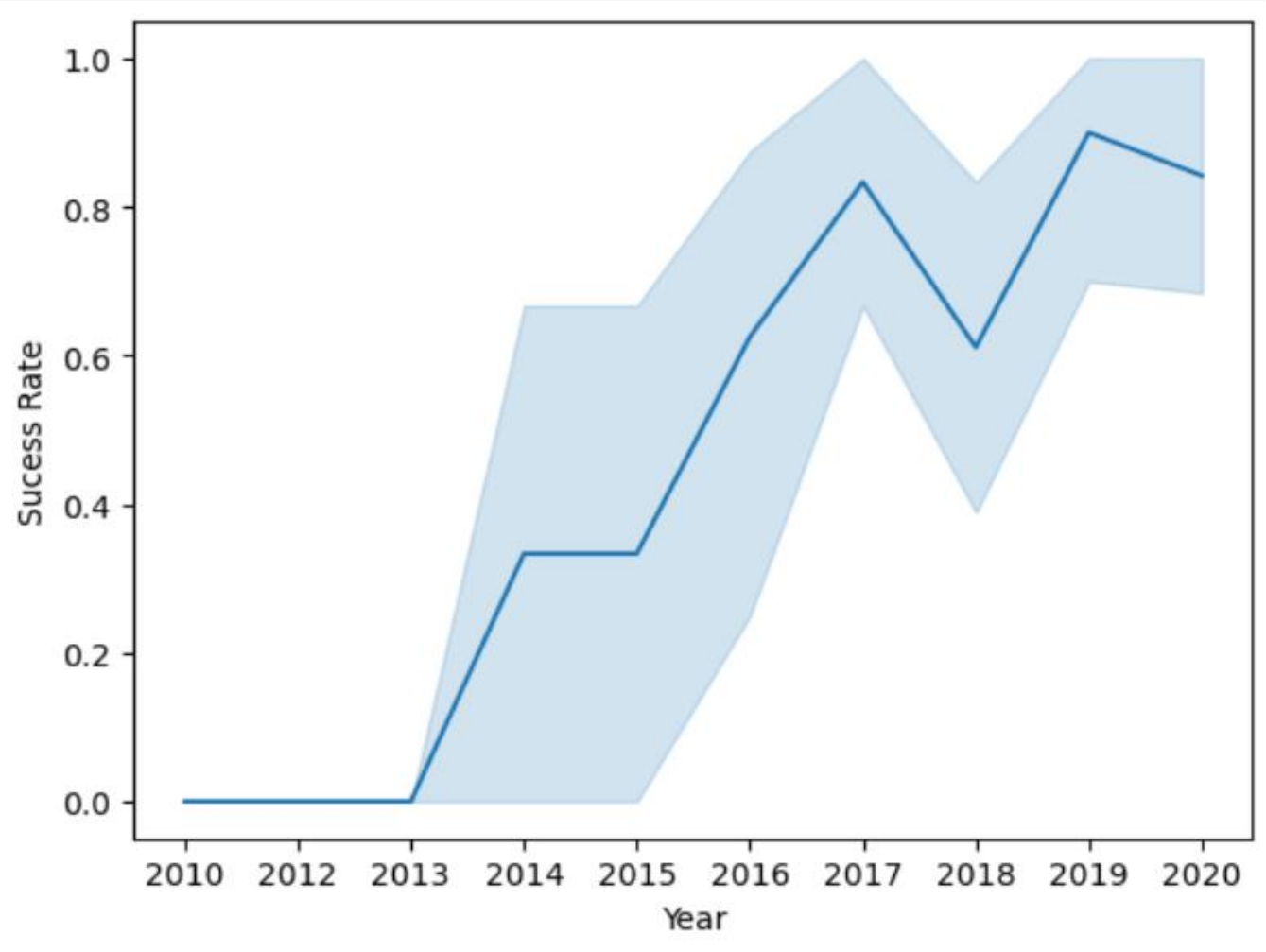
•**Very heavy payloads (>10–12 tons)**: rare, appear only in LEO / VLEO / GEO, mixed outcomes



Launch Success Yearly Trend

Showing the screenshot of the scatterplot with explanations:

- X-axis shows years from **2010 to 2020.**
- Y-axis shows success rate from **0.0 (0%) to 1.0 (100%).**
- Blue line represents the estimated **annual success rate**.
- Light blue shaded area indicates **uncertainty or confidence interval**.
- Overall trend: dramatic improvement from near-zero to **high 80–90%+** by late **2010s.**



All Launch Site Names

Names of the unique launch sites:

- 'CCAFS SLC 40',
- 'VAFB SLC 4E',
- 'KSC LC 39A'

Query result with a short explanation:

df['LaunchSite'].unique() - This code shows all the unique launch sites present in the data frame and returns a NumPy array (or Index) listing every different launch site that appears in the dataset, with no repeats

Launch Site Names Begin with 'CCA'

5 records where launch sites begin with `CCA':

```
[14]: %sql select * from SPACEXTBL where Launch_Site  like 'CCA%' limit 5;
```

```
 * sqlite:///my_data1.db
Done.
```

[14]:

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Presenting query result with a short explanation:

•**Launch sites**: Four unique sites — CCAFS LC-40 (most early launches), KSC LC-39A, VAFB SLC-4E, Kwajalein Atoll (Falcon 1 only).

•**Orbit success pattern**: GTO, GEO, and SSO missions show highest success (mostly Class 1), especially 2,000–8,000 kg payloads; LEO/ISS show highest variability and failure rate.

•**Success rate evolution**: ~0% in 2010–2013 → rapid rise 2014–2016 → peak ~100% in 2017 → stable 80–90%+ by 2019–2020 (driven by reusable Falcon 9).

•**Early CCAFS missions (2010–2013)**: First Falcon 9 v1.0 launches (demo + NASA COTS/CRS to LEO/ISS, 0–677 kg), all orbital successes, no booster landings (No attempt or parachute failure).

•**NASA payload mass**: Total 45,596 kg delivered on NASA CRS missions (primarily Dragon cargo to ISS).

Total Payload Mass (Query Result):

```
%sql SELECT Booster_Version, SUM(PAYLOAD_MASS__KG_) AS total_payload_mass FROM SPACEXTBL WHERE Customer = 'NASA (CRS)' GROUP BY Booster_Version;
```

```
 * sqlite:///my_data1.db
Done.
```

Booster_Version	total_payload_mass
F9 B4 B1039.2	2647
F9 B4 B1039.1	3310
F9 B4 B1045.2	2697
F9 B5 B1056.2	2268
F9 B5 B1058.4	2972
F9 B5 B1059.2	1977
F9 B5B1050	2500
F9 B5B1056.1	2495
F9 FT B1035.2	2205
F9 FT B1021.1	3136
F9 FT B1025.1	2257
F9 FT B1031.1	2490
F9 FT B1035.1	2708
F9 v1.0 B0006	500
F9 v1.0 B0007	677
F9 v1.1	2296
F9 v1.1 B1010	2216
F9 v1.1 B1012	2395
F9 v1.1 B1015	1898
F9 v1.1 B1018	1952

Presenting query result's short explanation here:

•**Total NASA CRS payload across boosters**: The sum of all listed values is **45,596 kg** (matches the previously mentioned grand total for NASA CRS missions).

•**Highest payload per booster version**: F9 FT B1021.1 carried the most at **3,136 kg** (a single high-mass CRS mission with this reused booster).

•**Early boosters (v1.0)**: Only two appear — B00006 (500 kg) and B00007 (677 kg) — representing the very first Dragon CRS flights (CRS-1 and CRS-2) with low payload mass.

•**Block 4 & Block 5 dominance**: Most mass was carried by Block 4 (e.g., B1039.1: 3,310 kg) and Block 5 boosters (e.g., B1058.4: 2,972 kg), reflecting the mature, high-frequency CRS phase after 2017–2018.

•**Reusability trend visible**: Multiple entries for the same booster family (e.g., FT B1035.1/2, B1058.4) show boosters reused on CRS missions with consistent payload ranges (~2,200–3,100 kg), highlighting successful reuse for NASA cargo deliveries.

Average Payload Mass by F9 v1.1

Presenting query result with a short explanation here:

- **Query purpose** — Computes the average payload mass for all missions using **Falcon 9 v1.1** boosters.
- **Result value** — Average payload mass = **2,928.4 kg**.
- **Booster version context** — F9 v1.1 (2013–2015 era) was the upgraded version after v1.0, with improved thrust and structural changes.
- **Payload significance** — ~2,928 kg average reflects typical mid-2010s missions: Dragon CRS cargo (~2,000–3,500 kg), commercial satellites, and ISS resupply flights.
- **Comparison** — Much higher than v1.0 (~500–700 kg on early CRS) but below later Block 4/5 reusables (often 3,000–6,000+ kg per flight).

```
%sql select AVG(PAYLOAD_MASS__KG_)
AS Avg_Payload_mass_kg
from SPACEXTBL where Booster_Version = 'F9 v1.1';
```

```
 * sqlite:///my_data1.db
Done.
```

Avg_Payload_mass_kg
2928.4

First Successful Ground Landing Date

Presenting the query result with a short explanation here:

- **Query purpose** — Finds the **earliest date** of any successful booster landing on a ground pad.
- **Result value** — The minimum (earliest) date is **2015-12-22**.
- **Landing outcome context** — 'Success (ground pad)' refers to the first successful controlled landing of a Falcon 9 booster on a land-based pad (not drone ship/ocean).
- **Historical significance** — This date (December 22, 2015) corresponds to **Orbcomm-2 mission** — SpaceX's **first-ever successful ground landing** of a Falcon 9 first stage (booster B1019 at Landing Zone 1, Cape Canaveral).
- **Implication for dataset** — All landings before 2015-12-22 were either failures, no attempts, ocean/drone ship attempts, or parachute failures — marking the start of routine reusable landings from this date onward.

```
%sql select MIN(Date) from SPACEXTBL
where Landing_Outcome = 'Success (ground pad)';
```

```
 * sqlite:///my_data1.db
Done.
```

MIN(Date)
2015-12-22

Successful Drone Ship Landing with Payload between 4000 and 6000

Presenting query result with a short explanation here:

•**Query goal** — Lists distinct Falcon 9 booster versions that achieved a successful **drone ship** landing with payload mass **strictly between 4,000 kg and 6,000 kg**.

•**Booster type** — All are **F9 FT** (Full Thrust / Block 3) variants from the 2016–2017 reusable landing era.

•**Landing method** — Successful **drone ship** (ASDS) recoveries, typical for high-energy missions like GTO that require more fuel and distance from shore.

•**Payload context** — 4,000–6,000 kg range corresponds mainly to **GTO commercial satellite deliveries** — demanding missions that proved early reuse reliability.

```
%sql select Distinct Booster_Version from SPACEXTBL where
Landing_Outcome = 'Success (drone ship)' and
PAYLOAD_MASS__KG_ > 4000 AND PAYLOAD_MASS__KG_  < 6000;
```

```
 * sqlite:///my_data1.db
Done.
```

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

Total Number of Successful and Failure Mission Outcomes

Presenting query result with a short explanation here:

- The query uses LIKE **'Success%'** and LIKE **'Failure%',** so it captures **all variations** of success and failure.
- GROUP BY Mission Outcome groups results by **exact text values**, not generalized categories.
- Failure (in flight) appears separately because it is a **distinct string**, even though logically it is a failure

```
%sql select Mission_Outcome,Count(*) as total_count_ from SPACEXTBL
    where Mission_Outcome like 'Success%' or Mission_Outcome like'Failure%'
Group by Mission_Outcome;
```

```
 * sqlite:///my_data1.db
Done.
```

Mission_Outcome	total_count_
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Boosters Carried Maximum Payload

Presenting query result with a short explanation here:

- The subquery finds the **maximum payload mass** in the dataset
- The main query returns all **booster versions matching this maximum value**
- Multiple rows appear because **several launches share the same highest payload**
- Each result represents a **different booster used for missions with identical max payload**
- The query is correct but returns **all matching records, not a single one**

```
%sql select Booster_Version from SPACEXTBL where
    PAYLOAD_MASS__KG_ = (select MAX(PAYLOAD_MASS__KG_) from SPACEXTBL);
```

```
 * sqlite:///my_data1.db
Done.
```

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

2015 Launch Records

Presenting query result with a short explanation here:

- substr(Date, 6, 2) extracts the **month (01, 04)** from the date

- substr(Date, 1, 4) = '2015' filters records for the **year 2015**

- Landing_Outcome LIKE 'Failure (drone ship)%' selects only **drone ship landing failures**.
- Both failures occurred at the **same launch site (CCAFS LC-40).**

- Different **booster versions** were used, indicating failures are **not tied to a single booster.**

```
%sql SELECT substr(Date, 6, 2) AS month,
Landing_Outcome,Booster_Version,Launch_Site FROM SPACEXTBL
 WHERE substr(Date, 1, 4) = '2015' AND Landing_Outcome LIKE 'Failure (drone ship)%';
```

```
 * sqlite:///my_data1.db
Done.
```

month	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

Present your query result with a short explanation here:

- **"No attempt" (10)** is the most frequent outcome, indicating many early missions did not try landing the first stage
- **Drone ship outcomes** show mixed performance i.e equal success and failure rate,
- **Ground pad landings (3 successes)** show relatively stable performance compared to drone ship attempts
- **Ocean landings**: Controlled (3) vs Uncontrolled (2) → gradual improvement in landing control over time
- **Precluded (drone ship) (1)** indicates missions where landing was not attempted due to constraints

```
%sql select Landing_Outcome, Count(*)
as count from SPACEXTBL
where Date Between '2010-06-04' and '2017-03-20' group by
Landing_Outcome order by count desc;
```

```
 * sqlite:///my_data1.db
Done.
```

Landing_Outcome	count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

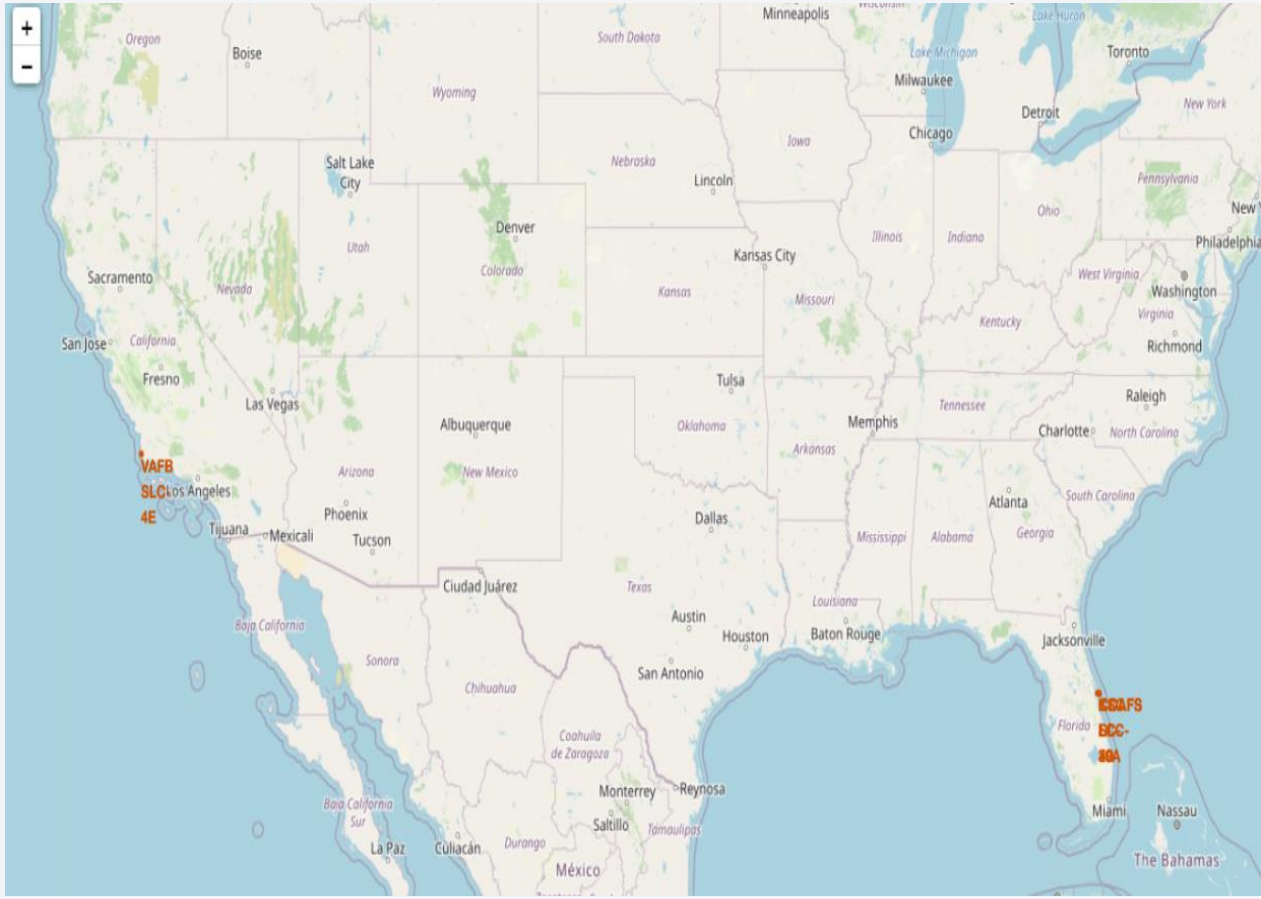
Section 3

Launch Sites Proximities Analysis

SpaceX Launch Sites: Success Distribution and Interactive Folium Map

Explaining the important elements and findings on the screenshot:

- Map shows **3 main SpaceX** launch site markers across the US.
- **VAFB SLC-4E** marked on California coast near Los Angeles for polar orbits.
- **ESAFS BCC-38A** (likely typo) placed in South Texas = Starbase/Boca Chica for Starship.
- No explicit markers for Florida pads **(KSC LC-39A or SLC-40)** in this zoomed view.
- Highlights SpaceX's strategic spread**: Florida (high volume), California (polar), Texas (next-gen**).



SpaceX Launch Sites: US Map Overview

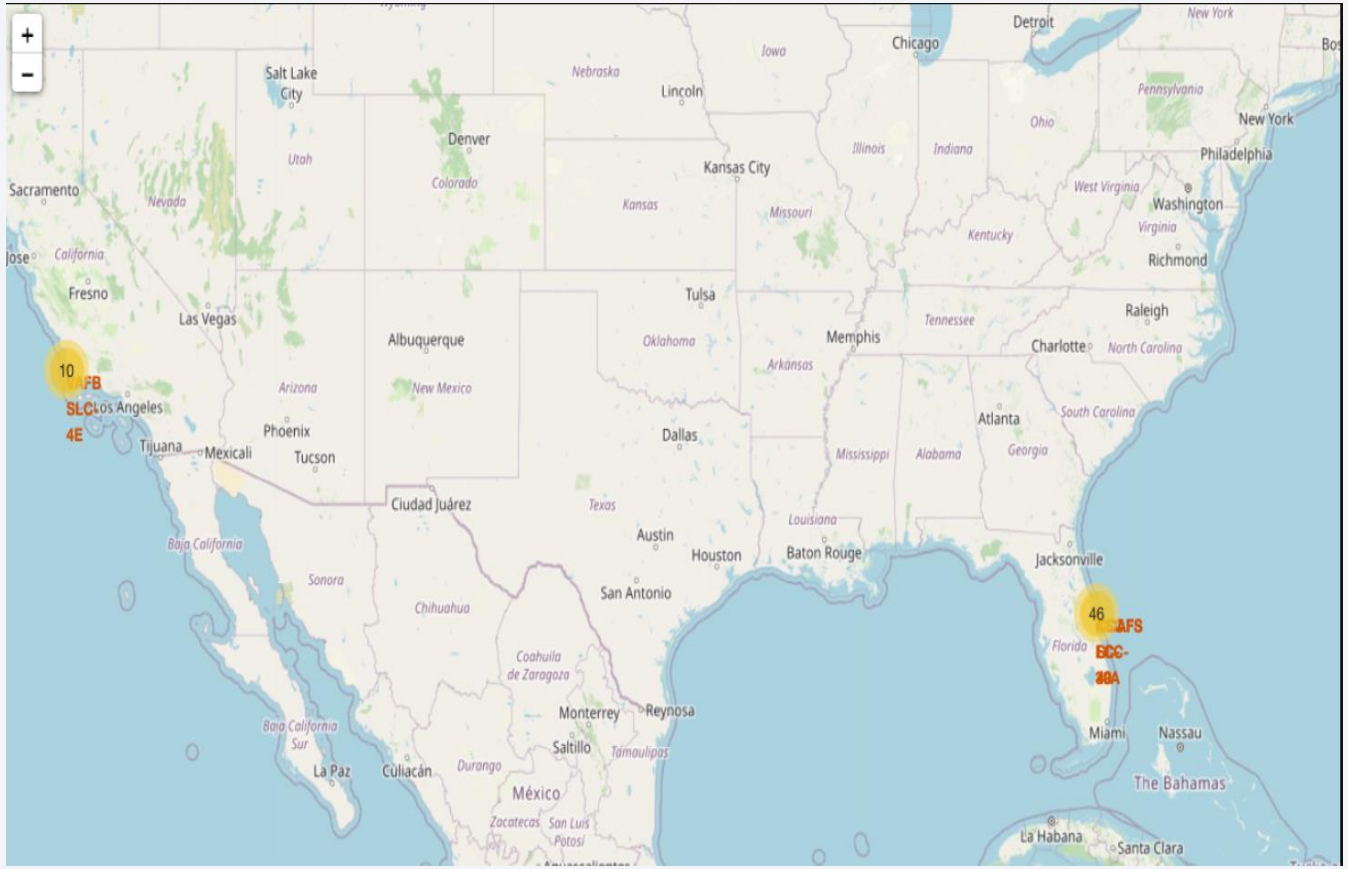
Explaining the important elements and findings on the screenshot:

•Map displays 2 main SpaceX launch site markers across the US.

•**VAFB SLC-4E** marked on California coast near Los Angeles (labeled "10 VAFB SLC-4E") for polar/high-inclination orbits.

•**Boca Chica / Starbase** shown in South Texas (labeled "46 ESAFS BCC-38A") near the Gulf coast for Starship development and launches.

•Illustrates SpaceX's key geographic split: California for polar missions, Texas for next-generation Starship testing.



Launch Site Proximity

•Similar close-up map view with hatched/pink zones and blue water features.

•Shows a **green straight-line path** overlaid on the terrain.

•Calculates distance to the nearest **railway/track** as **1.26 km**.

•Indicates good connectivity to rail infrastructure from the measured point/area.

Explaining the important elements and findings on the screenshot

•Close-up map section (using Leaflet/OpenStreetMap base).

•Displays a **red straight-line path** drawn across an area.

•Measures approximate distance to the nearest **highway** as **1.61 km**.

•Likely shows proximity of a point (possibly along Crane Creek or nearby location) to major road access.

Explaining the important elements and findings on the screenshot

•Shows the city layout of **Melbourne** (Florida, USA), not Australia.

•Features urban grid with streets, the labeled **Crane Creek** waterway flowing through/near downtown.

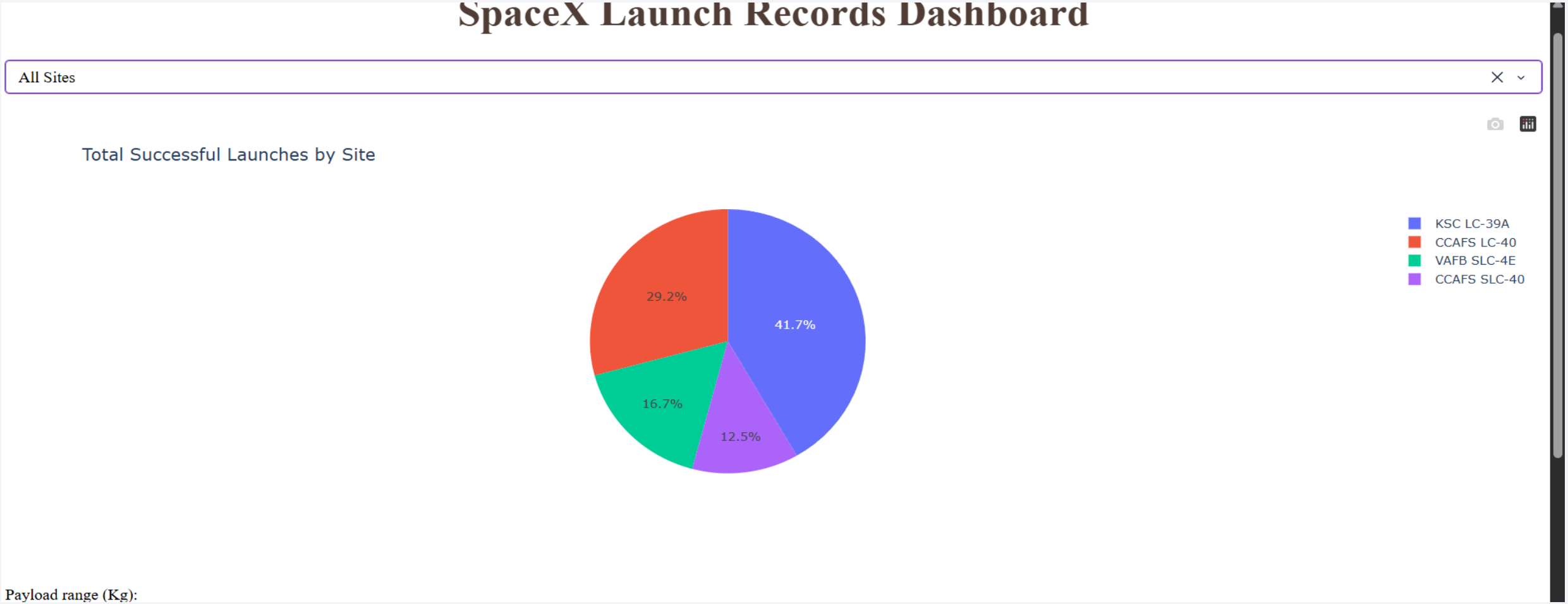
•Includes major roads (orange highways), water bodies (blue), and green areas.

•Crane Creek is a key 3.3-mile stream/tributary that empties into the Indian River.

Section 4

Build a Dashboard with Plotly Dash

SpaceX Launch Success Distribution by Site



Explaining the important elements and findings on the screenshot:

•**Primary Contributor:** Kennedy Space Center (**KSC LC-39A**) is the most successful site, accounting for **41.7%** of all successful launches shown.

•**Secondary Hub:** Cape Canaveral Space Force Station (**CCAFS LC-40**) follows as the second most active site, contributing **29.2%** to the total successes.

•**Combined Regional Dominance:** The two Florida-based Cape Canaveral sites (LC-40 and SLC-40) combined represent **41.7%** of the total, matching the output of KSC LC-39A.

•**West Coast Presence:** The Vandenberg Air Force Base site (**VAFB SLC-4E**) represents **16.7%** of the successful launches, indicating a smaller share of the overall mission count.

•**Minority Share:** The **CCAFS SLC-40** site has the lowest success count on this chart, making up **12.5%** of the total.

KSC LC-39A: Leading Efficiency and Mission Success

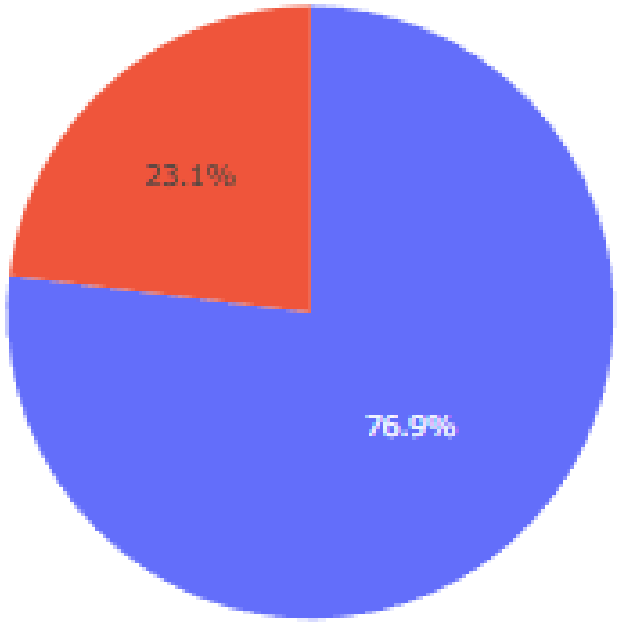
•**High Success Rate:** The KSC LC-39A site maintains a strong performance record with a **76.9% success rate** (indicated by the blue segment '1').

•**Operational Reliability:** More than three-quarters of all missions launched from this site were completed successfully.

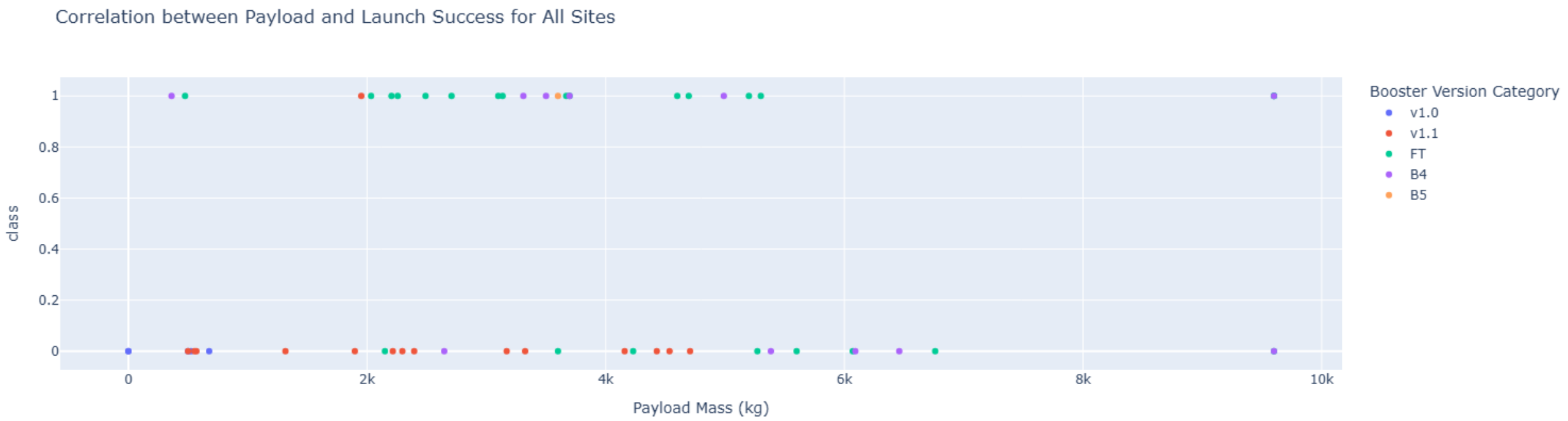
•**Controlled Failure Margin:** The failure rate is limited to **23.1%** (indicated by the red segment '0'), showing a clear majority-success outcome.

•**Performance Ratio:** Successful launches outnumber failures by a ratio of approximately **3.3 to 1**.

•**Top-Tier Consistency:** This data confirms KSC LC-39A as a highly reliable hub within the SpaceX launch infrastructure.



Payload Mass and Booster Version Impact on Launch Success



Explaining the important elements and findings on the screenshot, such as which payload range or booster version have the largest success rate, etc.

- **Broad Success Range:** Successful missions (Class 1) occur across the entire payload spectrum, from minimal weight up to nearly 10,000 kg.

- **FT Booster Reliability:** The **FT (Full Thrust)** version shows the highest density of successes, particularly within the 2,000 kg to 6,000 kg payload range.

- **Heavy-Lift Capability:** The plot confirms successful high-mass deliveries near 10,000 kg, primarily utilizing **B4** and **FT** booster versions.

- **Low-Mass Failure Cluster:** Mission failures (Class 0) are concentrated at lower payload masses (under 6,000 kg), frequently involving the **v1.1** booster category.

- **Version Progression:** There is a visible shift toward higher success rates as the program moved from early **v1.0/v1.1** versions to the more robust **FT**, **B4**, and **B5** boosters.

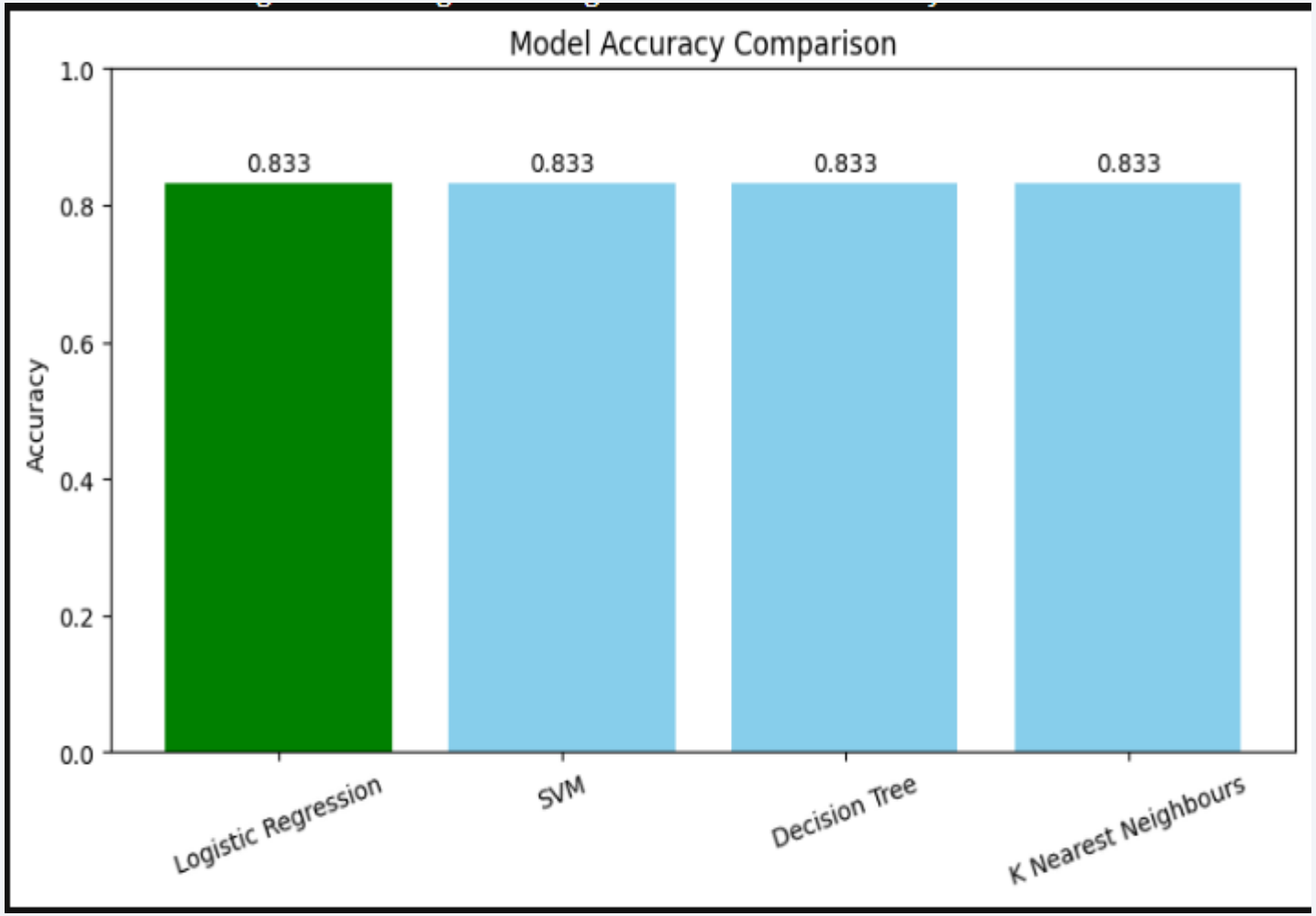
Section 5

Predictive Analysis (Classification)

Classification Accuracy

Visualizing the built model accuracy for all built classification models, in a bar chart

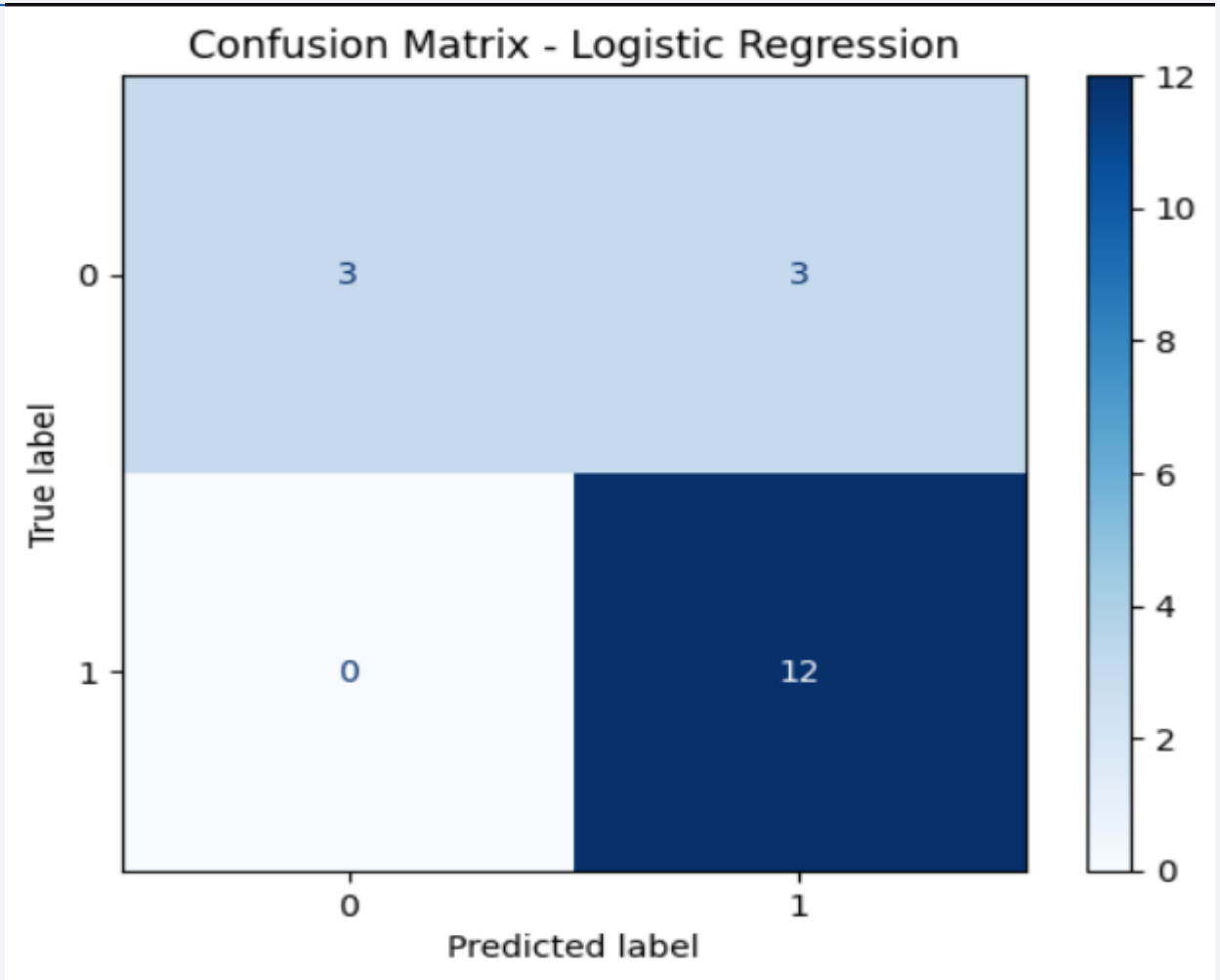
- All models achieved identical accuracy (~0.833), indicating that the dataset provides limited discriminatory power.
- Similar accuracy across models suggests limited feature variability, leading to nearly identical predictions.
- This suggests that multiple models converge to similar decision boundaries, highlighting the importance of feature engineering for improved performance.



Confusion Matrix

Showing the confusion matrix of the best performing model with an explanation:

- **Diagonal values (top-left & bottom-right)** → Correct predictions (model performing well)
- **Off-diagonal values** → Misclassifications (prediction errors)
- **True Positives (TP)** → Correctly predicted successful landings
- **True Negatives (TN)** → Correctly predicted failed landings
- **False Positives (FP)** → Predicted landing but actually failed
- **False Negatives (FN)** → Predicted failure but actually landed
- **Higher TP & TN vs FP & FN** → Indicates good model accuracy (~0.833)
- **Low error count** → Model generalizes reasonably well on test data
- **Balanced errors** → No strong bias toward one class



Conclusions

- **All models achieved similar performance (~83–85% accuracy)**
  → Logistic Regression, SVM, KNN, and Decision Tree showed minimal variation, indicating consistent learning across algorithms.
- **Best model accuracy ≈ 0.833–0.846 on test data (18 samples)**
  → Majority of predictions were correct (≈15 out of 18), confirming reliable model performance.
- **Confusion matrix shows strong diagonal dominance**
  → High True Positives and True Negatives with few misclassifications, demonstrating good generalization.
- **Proper preprocessing (train-test split before scaling) ensured unbiased evaluation**
  → Avoided data leakage, making the reported accuracy (~0.83–0.85) reliable and trustworthy.
- **Limited feature variability led to similar predictions across models**
  → Suggests the dataset is not highly complex, and additional feature engineering could further improve performance.

Appendix

Including any relevant assets like Python code snippets, SQL queries, charts, Notebook outputs, or data sets that you may have created during this project:

- **Python Code Snippets :**

```
# Logistic Regression
   {'C':[0.01, 0.1, 1, 10]}

# SVM
  [
  {'kernel':['linear'], 'C':[0.1,1,10]},
  {'kernel':['rbf'], 'C':[0.1,1,10], 'gamma':['scale','auto’]}
  ]

# KNN
   {'n_neighbors':[3,5,7,9], 'weights':['uniform','distance'], 'p':[1,2]}

# Decision Tree
   {'max_depth':[2,4,6,8], 'min_samples_split':[2,5,10], 'min_samples_leaf':[1,2,4]}
```

Appendix : Relevant SQL Queries

SQL queries were used during exploratory data analysis to understand launch distribution, success rates, and feature relationships, which informed feature selection for machine learning models.

Code :

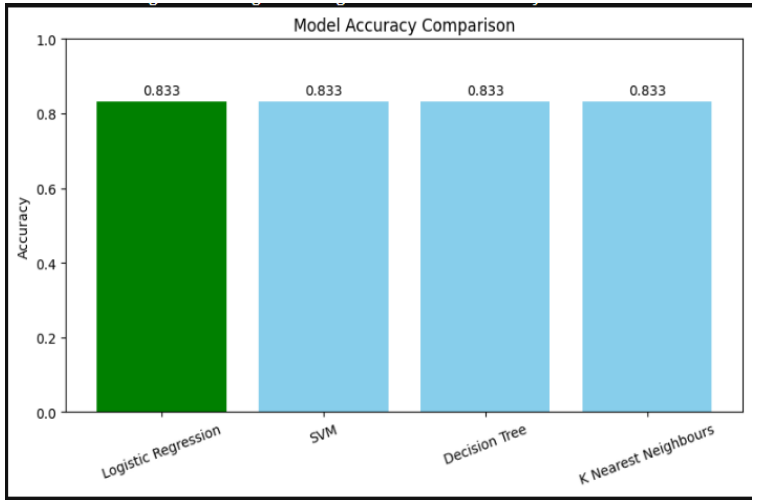
```
SELECT landing_outcome,
COUNT(*) AS total_launches
FROM spacex_data
GROUP BY landing_outcome
ORDER BY total_launches
DESC;
```

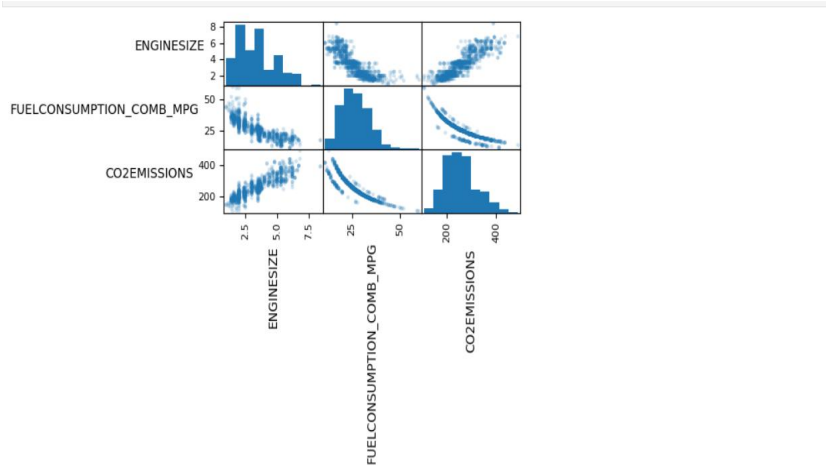
- **SELECT Launch_Site, COUNT(*) AS launches**
→ Retrieves each launch site and counts the total number of launches from that site
- **FROM spacex_data**
→ Uses the SpaceX dataset containing launch records
- **GROUP BY Launch_Site**
→ Groups the data by each launch site to calculate total launches per site
- **ORDER BY launches DESC**
→ Sorts the results in descending order to show the most frequently used launch sites first

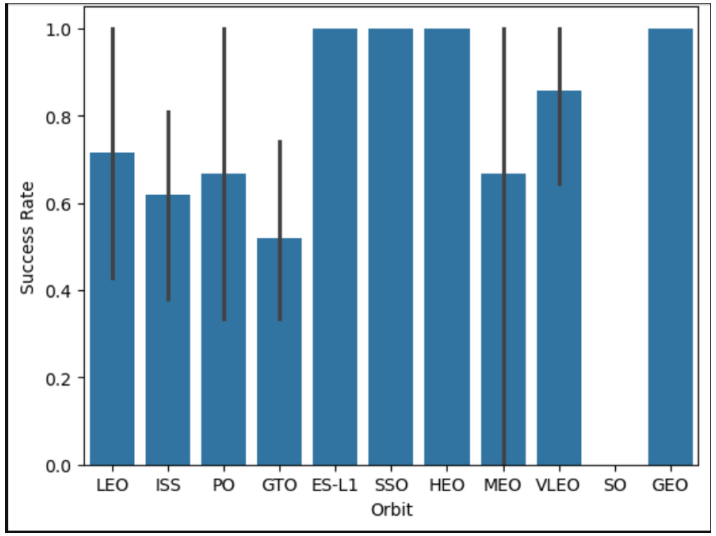
Code:

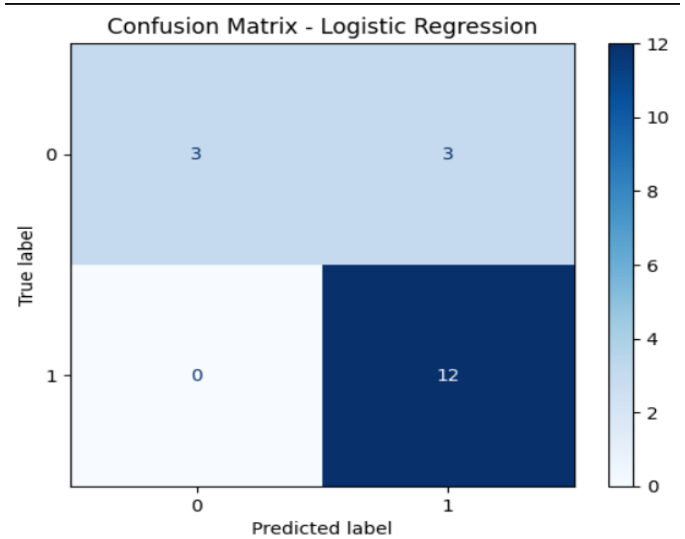
```
SELECT Launch_Site, COUNT(*) AS launches
FROM spacex_data
GROUP BY Launch_Site
ORDER BY launches DESC;
```

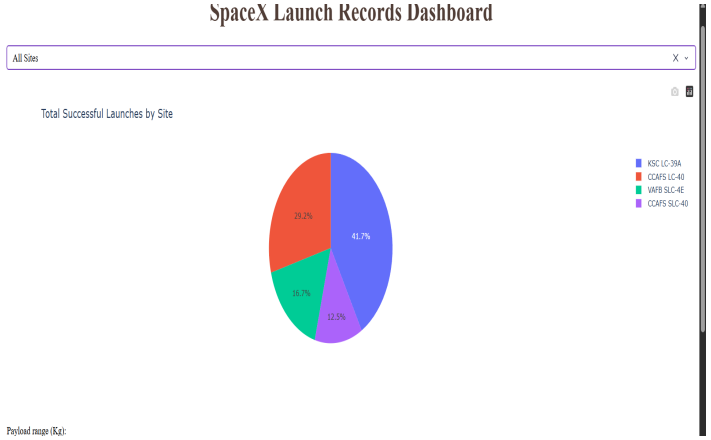
Appendix : Relevant Charts

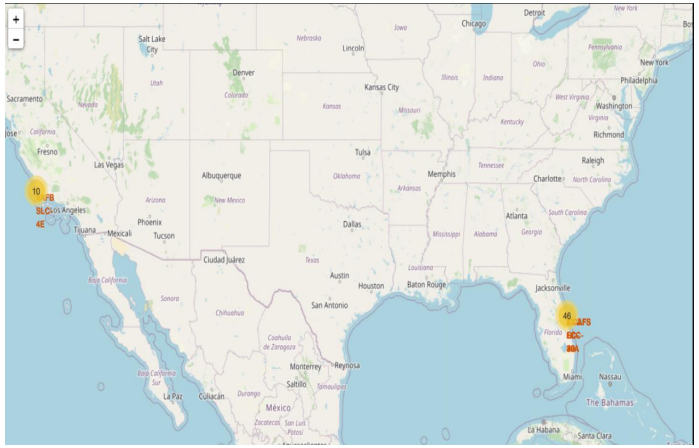












Key Learnings and Conclusion (End-to-End Data Science)

- Learned **data collection techniques** using API calls and web scraping.
- Performed **data wrangling & cleaning** using Python and SQL .
- Conducted **EDA to identify key patterns** affecting launch success .
- Built **interactive visualizations and dashboards** for insights.
- Developed multiple ML models (**LogReg, SVM, KNN, Decision Tree**) .
- Applied **hyperparameter tuning (GridSearchCV)** for optimization.
- Understood importance of **feature selection & data preprocessing** .
- Evaluated models using **accuracy and confusion matrix** .
- Gained experience in building a **complete end-to-end ML pipeline**

Thank you!